Tilt switch

DETECT PICO INPUT

What does it do?

A tilt switch changes state when it tips past a particular angle.

While it stands upright, a tiny metal ball rests on two contacts and completes the circuit. Tip it over and the ball rolls away, breaking the circuit.

It can detect that an object has been picked up, tipped over or opened.

Uses



Iron safety shut-off

DETECT: Iron tips sideways → DECIDE: Has the delay elapsed? → DO: Switch the heater off



Grumpy pencil case

DETECT: Case is tipped → DECIDE: Has it stayed tipped? → DO: Make a grumpy noise



Metal can
Contains the rolling ball

Legs
No polarity

Angle
Changes the contact

Raspberry Pi Foundation

Code

Raspberry Pi Pico

```
from picozero import Switch

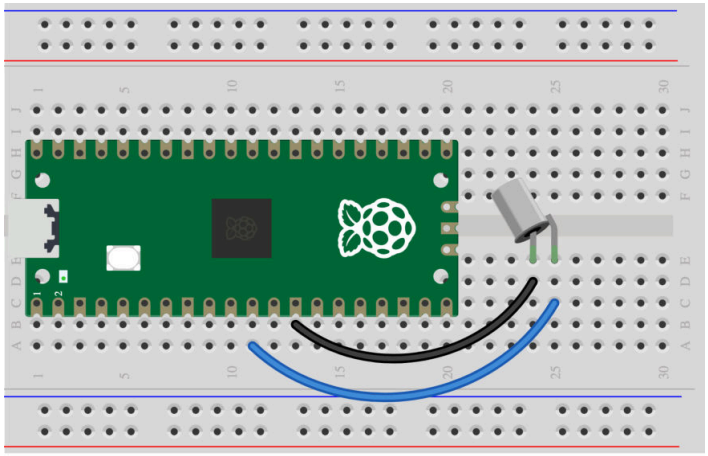
tilt = Switch(8)

tilted = tilt.is_open
```

> The reading may flicker while the ball rolls. Check it more than once before treating it as a definite tilt.

Wiring

PICO



GP8 (pin 11) → tilt switch → **GND** (pin 13)

Raspberry Pi Foundation